🏆 HACKATHON PROBLEM BOOKLET

Xavathon 2026

🎯 Organized By

Department of Computer Science
St. Xavier's College, Ranchi

📅 28–29 April 2026

📍 Proost Hall, St. Xavier's College, Ranchi

🏛 Theme

Innovation • Technology • Real-World Solutions

✨ About the Event

This problem booklet brings together a curated set of real-world challenges designed to inspire innovation and creativity. Participants are encouraged to build practical, scalable, and user-centric solutions using modern technologies.

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Think Smart. Build Fast. Solve Real Problems.

MindMate AI – Your 24/7 Mental Health Companion

Problem Statement

Mental health issues such as stress, anxiety, depression, and loneliness are rapidly increasing among students and young individuals. However, a large portion of people do not seek help due to high therapy costs, lack of immediate access to professionals, social stigma, and fear of judgment.

Challenge

Design an AI chatbot that delivers 24/7 emotional support, understands user mood through conversation, provides personalized coping strategies, and safely detects and escalates high-risk mental health situations.

Expected Features

1. Smart Conversational AI (Core Engine)
2. Mood Detection & Tracking
3. Daily Journaling
4. Personalized Suggestions
5. Crisis Detection System

Suggested Technologies

1. React / Next.js
2. Node.js / Express.js
3. OpenAI API (for chatbot intelligence) / Gemini API
4. MongoDB (for storing chats & mood data)
5. Tailwind CSS (for UI design)

Impact

This project democratizes mental health support by making it instant, affordable, and stigma-free for everyone.

FinTrack - Turn Every Expense into Insight

Problem Statement

Most individuals, whether they have regular or irregular income, struggle to manage their finances due to lack of proper tracking and clear insights. They are unable to understand where their money is being spent, which leads to overspending and poor financial planning. As a result, they find it difficult to save and achieve their financial goals

Challenge

Build a user-friendly finance dashboard that enables individuals to track income and expenses, visualize spending patterns through clear charts, and manage budgets effectively to improve financial planning and savings.

Expected Features

1. Expense & Income Tracking
2. Category-wise Spending Management
3. Analytics Dashboard (Charts & Insights)
4. Budget Planning & Control
5. Monthly & Weekly Financial Reports

Suggested Technologies

1. React / Next.js
2. Node.js / Express.js
3. MongoDB (for storing transactions & user data)
4. Chart libraries (Chart.js / Recharts for analytics dashboard)

Impact

Empowers users to take control of their finances through clear insights, better budgeting, and smarter spending decisions.

CampusKart - Buy Smart. Sell Easy. Right on Campus.

Problem Statement

Many students in colleges want to buy affordable second-hand items such as books, gadgets, and essentials, while others want to sell items they no longer need. However, there is no dedicated, trusted platform within the college to connect buyers and sellers. As a result, students miss out on affordable options and easy selling opportunities within their own campus.

Challenge

Build a secure and easy-to-use campus marketplace platform that enables students to buy and sell second-hand items within their college, ensuring trusted interactions, simple listings, and smooth communication between buyers and sellers.

Expected Features

1. User Registration & College Verification
2. Item Listing (Upload photos, price, description)
3. Browse & Search Products (category-wise)
4. Chat System between Buyer & Seller
5. Wishlist / Save Items Feature

Suggested Technologies

1. React / Next.js
2. Node.js / Express.js
3. MongoDB (for users, items, orders, chats)
4. Real-time communication (WebSockets / Socket.IO for chat)

Impact

Enables a smart, affordable, and sustainable campus economy by connecting students to buy and sell within their community

SteganoSafe – Secure Image-Based Communication

Problem Statement

In today's digital world, transmitting confidential information over the internet is highly vulnerable to hacking and unauthorized access. Traditional encryption methods can attract attention, making sensitive data a target for attackers. Therefore, there is a need for a secure and less detectable method of communication that not only protects the data but also hides its existence

Challenge

Build a secure system that allows users to hide encrypted messages inside images using a private key and safely retrieve them only with the correct key.

Expected Features

1. Image Upload & Message Input
2. Private Key-Based Encryptio
3. Data Embedding using LSB Technique
4. Secure Message Extraction & Decryption
5. User-Friendly Web Interface

Suggested Technologies

1. Python + HTML / CSS (Web)
2. Flask
3. OpenCV (image processing) + NumPy (pixel manipulation)
4. Pillow (image handling) + Cryptography

Impact

Enhances secure communication by combining encryption and steganography, making sensitive data both protected and undetectable during transmission.

GestureSense – Hand Gesture Recognition System

Problem Statement

Traditional input methods like keyboards and touchscreens are not always convenient or accessible in every situation. There is a growing need for natural and touchless interaction systems that allow users to communicate with devices more intuitively. However, existing solutions are often complex, expensive, or not easily accessible. Therefore, a simple and efficient system is required to recognize hand gestures in real time and convert them into meaningful outputs.

Challenge

Build a real-time system that detects and recognizes static hand gestures using a webcam and converts them into text output for intuitive human-computer interaction.

Expected Features

1. Real-Time Hand Detection using Webcam
2. Static Gesture Recognition (10+ gestures)
3. Machine Learning-Based Classification
4. Text Output for Recognized Gestures
5. Simple & User-Friendly Interface

Suggested Technologies

1. Python
2. OpenCV + MediaPipe / CVZone
3. NumPy + Scikit-learn
4. Webcam (for real-time input)

Impact

Provides a simple and accessible way for touchless interaction, useful in scenarios like accessibility for disabled users, smart interfaces, and hygienic environments.